

What Does a Yes/No LLM Recommender’s Confidence Know? Item Priors, Personal Evidence and Actionable Uncertainty

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Abstract

[PENDING: abstract]

CCS Concepts

• Information systems → Recommender systems.

Keywords

LLM-based recommendation, confidence and calibration, item prior, popularity bias, selective prediction

[PENDING: introduction]

1 Related Work

Pointwise yes/no scoring and its confidence. TALLRec [5] tunes an LLM with LoRA to answer whether a user will like an item, CoLLM [76] and BinLLM [74] add collaborative information, and ReLLa [44] retrieves behaviours and scores a pair by the two-way Yes/No softmax, the sigmoid of the logit audited here. These models are judged by discrimination (AUC, UAUC, log loss), rarely by reliability. IR rankers score relevance by the probability of a “true” token [52]; finer relevance grades help zero-shot pointwise rankers [83], and pairwise prompting avoids the calibration across prompts that pointwise scores need [55]. Generative recommenders [3, 18, 42] face distorted sequence-level confidence [4], which one Yes/No position avoids; zero-shot LLM rankers are swayed by popularity and candidate position [26], and LLMs partly memorise MovieLens-1M [12]. At the output level, LLM recommenders recover personal preference poorly under pointwise rating prompts once user rating bias and item quality are controlled [61], zero-shot LLMs trail collaborative filtering in rating prediction while fine-tuned ones approach it [32], and they barely use the order of the history [34]. None of these measures what the token confidence itself encodes, how fine-tuning changes it, or which uses of it survive a matched control.

Uncertainty, calibration and answer priors. LLMs are overconfident when they verbalise confidence [69] but fairly calibrated in token probabilities on multiple-choice questions [31]; fine-tuning shapes confidence [33, 41], and calibration can depend on how close an input is to others [68]. Binary answers carry input-independent priors, removed by contextual [79] or batch calibration [81] or PMI scoring [24]; yes/no answers carry order and wording biases that crossing a statement with its negation separates, leaving idiosyncratic per-item residuals [28], and LLMs prefer recognised brands [10]. The two-view contrast evaluated here is the log-odds analogue of the symmetrised inference rule of contrast-consistent

search [8]. For LLM recommenders, UQRec [37] decomposes list-wise uncertainty, GUIDER [70] re-ranks by data and model uncertainty, UGR [13] uses confidence tokens for rejection and truncation, EviRank [71] estimates position-level confidence, an audit scores verbalised confidence by popularity stratum [56], KnowSA [40] finds popularity and uncertainty poor proxies of item knowledge, uncertainty-aware recommendation adapts to low-active users [84], FACTER thresholds a fairness score conformally [14], and selective escalation to an LLM did not help in a sparse study [82]. Calibrating personalised ranking scores is older [38]. The decomposition below is the biased-MF baseline predictor applied to log-odds, with the item term read off the model, and echoes the separation of interest from conformity in causal embeddings [80]; a prior-offset LoRA is a product of experts with a bias-only expert [11].

Popularity, exposure, pruning and serving. Popularity bias [1] has causal treatments [66, 75] and is amplified by SFT and DPO in LLM recommenders [15, 16], whereas a prompted LLM recommender showed less of it than conventional ones [43] and LLM recommenders can be unfair to items [29]; in entity question answering, LLM confidence rises with answer popularity [51]. Feedback loops raise homogeneity and popularity bias [9, 49], also with LLM components [53]; we measure exposure statically. LLM-recommender training data are selected by influence and effort [45] or by core-sets, where loss- and gradient-based scores fall below random [50] (neither tests the uncertainty of a zero-shot model, so we compare with class-matched random pruning); training dynamics and data pruning more generally [58, 60], uncertainty-aware instruction selection [46] and claim-level cuts at an uncertainty quantile [67] are related, and recommenders are denoised by dropping large-loss interactions [63] or by thresholding an LLM’s generation probability [62]. Serving by a confidence score is a performance-prediction decision [6] and selective classification [17], which can widen group gaps [30]. We test each use against a matched control.

2 Setting and Notation

Confidence of a yes/no recommender. A pointwise yes/no recommender [5, 76] scores a user u and a candidate item i by asking a binary question q about the pair (here “Would this user like the candidate item?”) and reading the next-token distribution of a language model f , zero-shot or with a LoRA adapter fitted to yes/no labels. Its *confidence* is the log-odds

$$\ell(u, i) = \log P_f(\text{Yes} \mid x_q(u, i)) - \log P_f(\text{No} \mid x_q(u, i)), \quad (1)$$

where $P_f(\text{Yes} \mid x)$ sums the probabilities of the vocabulary ids that decode, stripped and lower-cased, to “yes” (likewise No). Candidates are ranked by ℓ .

Panels and metrics. A *unit* is a user (rated panels) or a prediction event (next-item panels) with candidate set C_u and labels $y(u, i) \in \{0, 1\}$. *Rated panels* show the user’s earlier ratings as history and use the user’s own later rated items as candidates, labelled 1 for a rating

of at least 4 and 0 for at most 2 (so the question matches the label). With $c(x) = 1, \frac{1}{2}, 0$ for $x > 0, x = 0, x < 0$, the per-user AUC of a score s and its mean over users with both classes (UAUC [74, 76]) are

$$\text{AUC}_u(s) = \frac{1}{|C_u^+||C_u^-|} \sum_{i \in C_u^+} \sum_{j \in C_u^-} c(s_i - s_j), \quad \text{UAUC} = \text{mean}_u \text{AUC}_u.$$

Same-candidate next-item panels hold, per event, one positive and 100 popularity-sampled negatives, ranked identically by every system (not full-catalogue ranking [36]); with a candidates above the positive and t tied with it, its rank is uniform on $1+a, \dots, 1+a+t$, and $\text{HR}@k$ and $\text{NDCG}@k$ are expectations over that rank ($k = 10$).

What the confidence is made of. Let $\pi(i) = \mathbb{E}_D[\ell(D, i)]$ be the *item prior*: the expected confidence for i under the history of a donor D who has not seen i , estimated by the mean $\hat{\pi}(i)$ over $K = 8$ donors. With the user offset $b_u = |C_u|^{-1} \sum_{i \in C_u} [\ell(u, i) - \pi(i)]$ the remainder is the *pair-specific residual* e ,

$$\ell(u, i) = b_u + \pi(i) + e(u, i), \quad \sum_{i \in C_u} e(u, i) = 0, \quad (2)$$

so within a unit only $\pi + e$ orders the candidates. This is the biased-matrix-factorisation baseline predictor applied to log-odds, with the item term read off the model itself. The residual is *not* personal evidence by construction: it also holds donor-population mismatch, user-by-prior interaction and history noise unrelated to the label. We therefore measure personal information by prediction. The *information gain* $\mathcal{G}$ is the UAUC gain of cross-fitted stackers when $\hat{e} = \ell - \hat{\pi}$ is added to $\hat{\pi}$ and an item mean $m(i)$ (other users' mean rating of i before the candidate's timestamp), and $\mathcal{G}_{\text{CF}}$ is the analogous gain of a matrix-factorisation (MF) residual over m . The *item-prior share* of the confidence is the share of its within-user variance that $\hat{\pi}$ explains, ρ^2/r_8 , with ρ the pooled within-user correlation of ℓ and $\hat{\pi}$ and r_8 the Spearman–Brown reliability of the 8-donor mean [7, 59]; it is reported only where $r_8 \geq 0.7$. Estimators are in Section 3 and Appendix ??.

Where can confidence change a ranking?

LEMMA 2.1 (INVARIANCE). *If every $g_u : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing and $s'_i = g_u(s_i)$ for $i \in C_u$, then $\text{sgn}(s'_i - s'_j) = \text{sgn}(s_i - s_j)$ for all $i, j \in C_u$; hence AUC_u , $\text{HR}@k$, $\text{NDCG}@k$ and MRR , which depend on s only through these signs, are unchanged.*

Temperature and Platt scaling [20, 54] with positive slope, removal of b_u and the list softmax are such maps: *per-user calibration cannot change UAUC or within-unit NDCG*; only non-decreasing maps (isotonic regression [73]) can, through ties. A correction that depends on the item, $s' = s - \lambda b(i)$, exchanges or ties a pair exactly when $s_i - s_j \leq \lambda(b(i) - b(j))$; contextual and batch calibration [79, 81] subtract a prior that is a constant of the unit for a binary question, whereas candidate-conditioned priors (the no-history prompt, in the spirit of PMI scoring [24]; the swap prior $\hat{\pi}$) are item-dependent. Across units the lemma is silent: offsets change pooled AUC, thresholds and selective serving, and training changes f . These are the routes left open, and Section ?? tests each against a matched control.

3 Audit Design

Panels, users and backbones. *Rated panels* (MovieLens-1M [21]; Amazon Reviews 2023 [25] Toys, Video Games, Sports) show up to ten earlier ratings of a user as history (at least three are required) and use the user's latest labelled items, at most 20, as candidates; a user needs three likes and three dislikes among them. Eligible users are shuffled with a fixed seed: the first 1,500 are TRAIN users, the next (up to 3,000) EVAL users, disjoint and never trained on. With T_d the 0.8-quantile of a panel's candidate timestamps, adapters train on TRAIN candidates before T_d , calibrators use EVAL candidates before T_d , and every endpoint is computed on EVAL candidates at or after T_d (TEST). *Next-item panels* are four frozen same-candidate panels of an earlier benchmark protocol [2] (Amazon Sports, Toys, Home, Tools; 10,000 events each, 101 candidates) with the top-10 lists of a verbalised reranker and eight published recommenders [22, 47, 48, 57, 64, 65, 77, 78] as context, scored zero-shot. The primary backbone is Qwen3-8B [72]; Llama-3.1-8B-Instruct [19] replicates on ML-1M and Toys (both regimes) and on next-item events 1,001–3,000. Scoring is one fp16 prefill pass in vLLM [39] reading Yes and No by token id; a run that fails the integrity check (answer tokens present, no over-length prompt, Yes+No mass at least 0.95) is rerun once and otherwise reported missing. The fine-tuned regime is TALLRec-style LoRA yes/no tuning [5, 27] with one fixed recipe (rank 16, one epoch, three seeds).

References on the same rows. Every rated table carries, on identical rows, the item mean $m(i)$ (other users' mean rating of i , shrunk, before the candidate's timestamp). m sees the whole rating corpus up to the candidate, far more than the LLM does, so it is the demanding reference; a *time-matched* mean m_T (ratings before T_d only) and the mean of the adapter's own TRAIN labels m_{TR} stand beside it. A temporal biased MF fitted before T_d [35] gives the collaborative reference, with its personal residual (MF $- b_u - b_i$); as registered it over-fits the three Amazon panels, so a regularisation-tuned MF restricted to the ten events the prompt shows, and a content-based personal reference that uses exactly the titles, categories and brands of the prompt, are added as exploratory references (Appendix ??).

Measures. Besides UAUC and the contrast with each reference, we report the item-prior share, the share of variance of $\hat{e}$, the information gain $\mathcal{G}$ with its registered pooled estimator and a within-user estimator that refits the stackers on user-centred features, and $\mathcal{G}_{\text{CF}}$ and $\mathcal{G}_{\text{LLM|CF}}$ (what the LLM residual adds to MF). Because $\mathcal{G}$ can be positive when ℓ merely denoises the 8-donor prior, three validity checks accompany it: a permutation null of $\hat{e}$ within users, the recovery of planted signals (power and minimum detectable gain), and a donor placebo that replaces the user's own confidence by a held-out donor's for the same item.

Training controls. Four adapter families remove user–item evidence from the training labels. *Permuted-label adapters* train on labels permuted within each item (ML-1M only: an item seen once cannot change); *item-teacher adapters* train on labels from the TRAIN item mean alone, so they use no user information; *item-balanced adapters* keep, for every item with both classes, equally many positive and negative examples, so the item label rate is 0.5 and the labels can be reached only through the user–item pair, with a size-matched random subsample and a within-item-permuted copy as

controls; a globally permuted adapter measures the readout and format shift. The retention R is the share of the LoRA gain over zero-shot that the control adapters keep.

Uses of the confidence. The lemma leaves four routes open, each tested against a matched control: cross-user *serving* and *routing* on the next-item panels (serve the most confident events, or send the others to a published recommender) against a random signal and cheap signals that need no LLM (history length, user popularity profile, the confidence of a popularity-only ranker); static *exposure* of the top-10 lists against the candidate pool and the published recommenders; item-dependent *corrections* against a paraphrase placebo and an ensemble null; and *pruning* of the LoRA training set by the zero-shot model's uncertainty against class-matched random pruning (five seeds). A prior-offset LoRA, which lets a learned offset absorb the item prior, is an optional method compared with post-hoc stacking.

Statistics, registration and what is claimed. Intervals are 95% percentile bootstraps over users or events (2,000 resamples, seed 0); contrasts are paired. Holm's correction [23] applies within registered families (the effect of fine-tuning over the four Qwen panels; the LLM-minus-item-mean contrast over the three Amazon panels; $\mathcal{G}$ per panel and regime; the method slot), a fine-tuned effect counts only if its seed mean exceeds twice the seed s.d. with all three seeds of one sign, and an endpoint on fewer than 150 users is descriptive. A statement beyond one panel is made only if it is confirmed on untouched users and replicates, with the same sign and its own interval excluding zero, on a fresh Amazon domain and on the second backbone; every other estimate is labelled descriptive. Each analysis was registered in a dated internal amendment whose SHA-1 hash was logged before it ran (the first amendment retroactively); the audit design followed a failed pilot (Appendix ??); the analyses written after the ML-1M report was read, and every later addendum, are exploratory where the results they concern had been seen and are set in italics. **[PENDING: findings]** **[PENDING: conclusion]** **[PENDING: appendix]**

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